Circles and Parabolas

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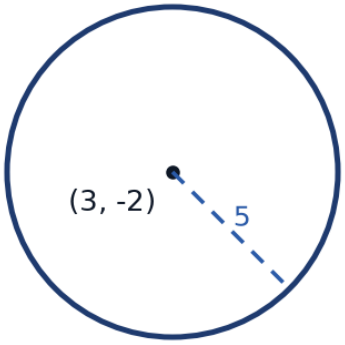
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Circles**

Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.

2 **[Foundational]** • **Target: Circles**

Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.



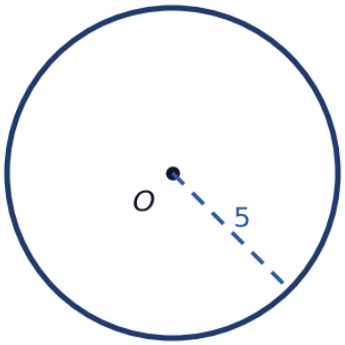
a Write the equation in standard form by completing the square.

b State the center and radius.

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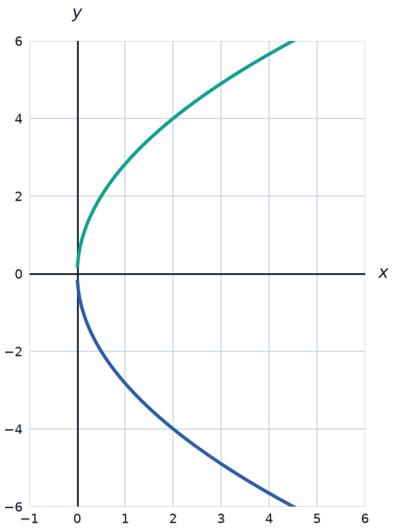
3 [Foundational] • Target: Circles

Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$.



4 [Foundational] • Target: Parabolas with vertex at the origin

For the parabola $y^2 = 8x$, find:



a the focus

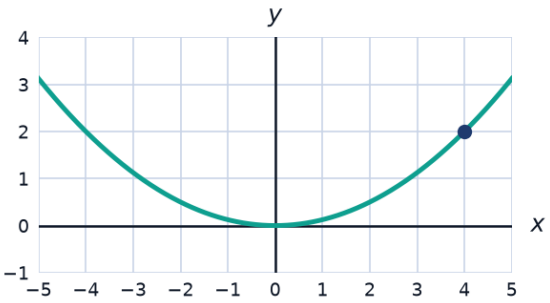
b the directrix

c the direction of opening

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5 [Foundational] • Target: Parabolas with vertex at the origin

A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.



a Find its equation in the form $x^2 = 4py$.

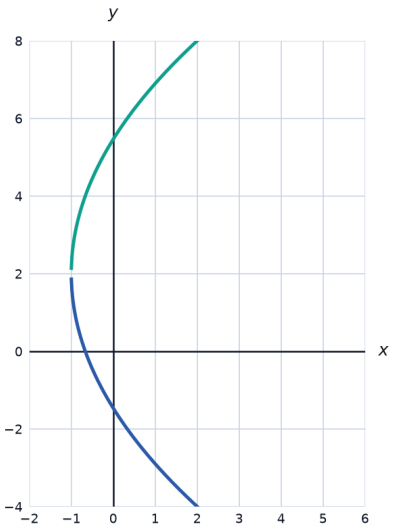
b State the coordinates of the focus.

6 [Foundational] • Target: Parabolas with vertex at the origin

A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).

7 [Foundational] • Target: Parabolas with vertex not at the origin

In a quantitative modeling scenario, for the parabola $(y - 2)^2 = 12(x + 1)$, find:



a the vertex

b the value of p

c the focus

d the directrix

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Parabolas with vertex not at the origin

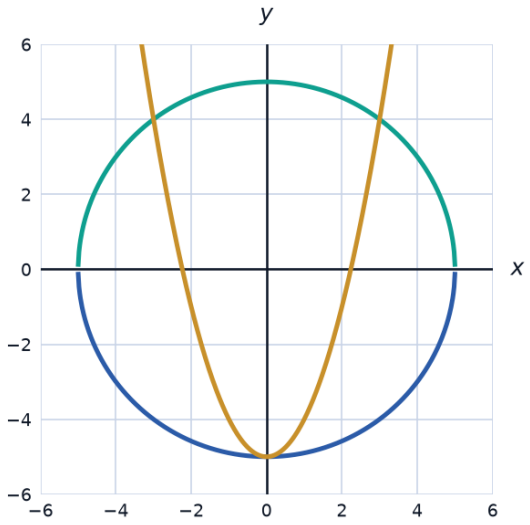
Write the equation of the parabola with vertex $(2,\ -3)$, opening downward, with $|p| = 1$.

9 [Intermediate] • Target: Parabolas with vertex not at the origin

Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.

10 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.

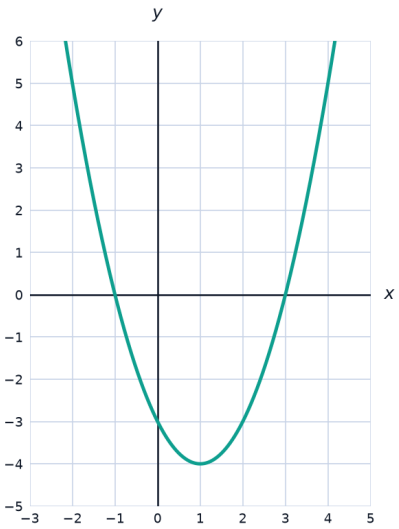


11 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.

12 [Intermediate] • Target: Graphing circles and parabolas

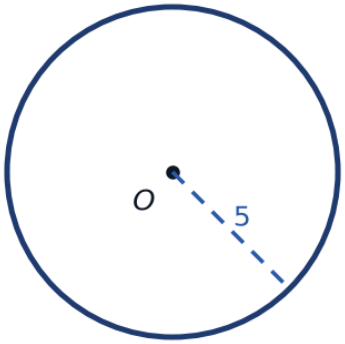
The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x-intercepts, then verify algebraically.



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13 [Intermediate] • Target: Tangent lines to circles

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.



14 [Intermediate] • Target: The reflective property of parabolas

Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

16 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 9.02 Geometric Sequences and Series

State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

17 [Challenging] • Target: Systems involving circles • Cross-Lesson: 9.03 The Binomial Theorem

Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$.

18 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 1.01 Functions and Function Notation

Find the equation of the circle in standard form that passes through the three points $(0, 0)$, $(6, 0)$, and $(0, 8)$.

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19 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 1.02 Domain and Range

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $P(3, 4)$.

20 [Challenging] • Target: Systems involving circles • Cross-Lesson: 1.03 Average Rate of Change

A tangent from an external point P touches a circle at T. If the distance from P to the circle's centre O is 13 and the radius is 5, find the length of the tangent segment PT.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Circles

A circle $\mathcal{C}$ with center $C(0, k)$ on the positive y-axis passes through the focus F of the parabola $x^2 = 4y$. If $\mathcal{C}$ is tangent to the parabola at two distinct points, P and Q, what is the area of quadrilateral $PFQC$?

a. $8\sqrt{3}$

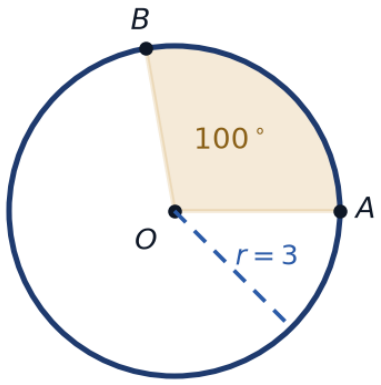
b. $4\sqrt{3}$

c. $16\sqrt{3}$

d. $6\sqrt{3}$

22 [Foundational] • Target: Parabolas with vertex at the origin

In the diagram below, a circle has its center at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the radius of the circle?



a. 5

b. 7

c. 25

d. 12

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23 [Foundational] • Target: Parabolas with vertex not at the origin

A parabola has its vertex at the origin $(0, 0)$ and passes through the point $(2, -4)$. If the directrix of the parabola is a horizontal line, what is the equation of the directrix?

a. y = 4

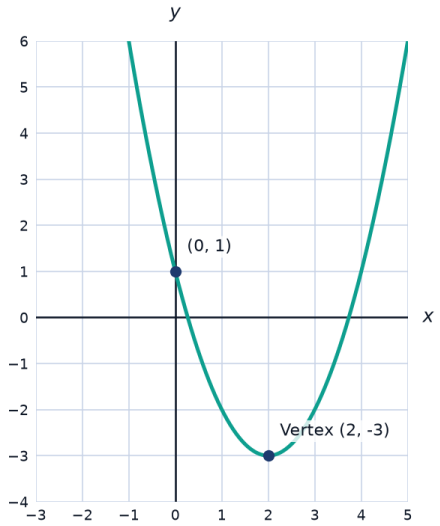
b. y = -4

c. x = 4

d. x = -4

24 [Foundational] • Target: Mixed practice: circles and parabolas together

The graph of the quadratic function $f(x) = x^2 - bx + c$ is shown below. The vertex of the parabola is at $(2, -3)$, and it passes through the point $(0, 1)$. Determine the value of the parameter b.



a. 4

b. 3

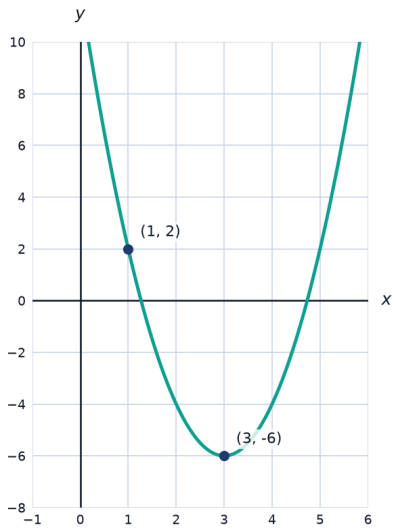
c. -4

d. -3

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25 [Foundational] • Target: Graphing circles and parabolas

The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b?



a. -12

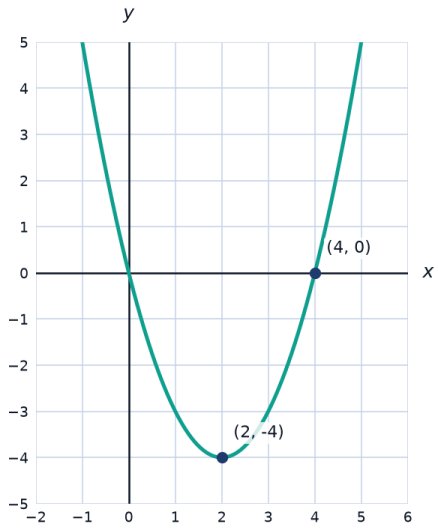
b. -6

c. 6

d. 12

26 [Foundational] • Target: Tangent lines to circles

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -4)$, and the function passes through the point $(4, 0)$. Determine the value of the coefficient a.



a. 1

b. -1

c. 2

d. -2

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27 [Foundational] • Target: The reflective property of parabolas

Consider the function $g(x) = ax^2 + bx + c$. The graph of $y = g(x)$ is a parabola that passes through the points $(1, 5)$, $(2, 12)$, and $(3, 21)$. If the vertex of the parabola is at (h, k), what is the value of h?

a. $\frac{5}{2}$

b. 2

c. $-\frac{5}{2}$

d. -2

28 [Foundational] • Target: Finding a circle through three points

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?

a. -7

b. 9

c. 12

d. 15

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Distance from a point to the focus (parabola definition)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

a. 1

b. 2

c. 3

d. 4

30 [Intermediate] • Target: Systems involving circles

A circle has center $(6, 6)$ and radius 2. Find the shortest distance from the point $(16, 30)$ to the circle.

a. 6

b. 38

c. 24

d. 29

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31 **[Intermediate]** • **Target: Circles**

A circle has center $(4, -2)$ and passes through $(13, 10)$. Find the slope of the tangent line to the circle at $(13, 10)$.

a. 5/4

b. 1/4

c. -3/4

d. -11/4

32 **[Intermediate]** • **Target: Parabolas with vertex at the origin**

Find all points on the parabola $y^2 = 4x$ equidistant from the focus and from the point $(6, 0)$.

a. $(4, 4)$ only

b. $(1, 2)$ only

c. $(0, 0)$

d. $(1, 2), (1, -2)$ and $(9, 6), (9, -6)$

33 **[Intermediate]** • **Target: Parabolas with vertex not at the origin**

Two tangent lines from an external point P to a conic: the segment from P to each tangent point has the property that the two tangent lengths are equal for a:

a. Hyperbola

b. Ellipse

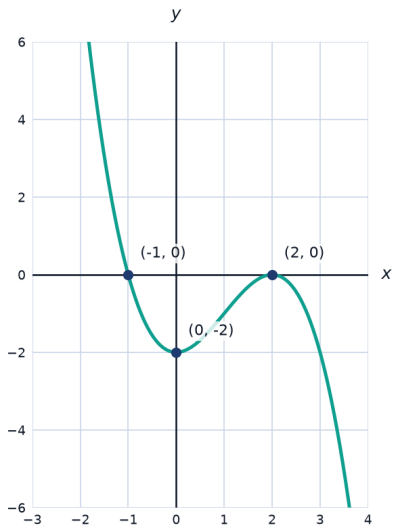
c. Circle

d. Parabola

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34 **[Intermediate]** • **Target: Mixed practice: circles and parabolas together**

The graph of a third-degree polynomial function $P(x)$ is shown in the standard (x, y) coordinate plane below. The graph crosses the x-axis at $(-1, 0)$, is tangent to the x-axis at $(2, 0)$, and passes through the point $(0, -2)$. What is the value of $P(-2)$?



a. -16

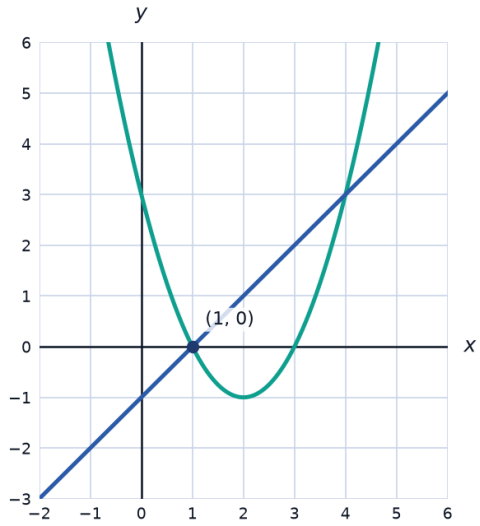
b. -8

c. 4

d. 8

35 **[Intermediate]** • **Target: Graphing circles and parabolas**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at two points. What is the y-coordinate of the vertex of this parabola?



a. -1

b. 1

c. -2

d. 0

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36 [Intermediate] • Target: Tangent lines to circles

In the diagram below, points A, B, C, and D lie on the circle. If AB = 6, BP = 8, CD = 4, and AP = x, where P is the intersection of chords AC and BD, find the value of x.

a. 4

b. 5

c. 6

d. 8

37 [Intermediate] • Target: The reflective property of parabolas

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

a. 3

b. -3

c. 6

d. -6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 1.01 Functions and Function Notation

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

a. $a = 1, b = -4, c = 3$

b. $a = 1, b = 4, c = -5$

c. $a = 2, b = -8, c = 7$

d. $a = -1, b = 4, c = -5$

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39 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 1.02 Domain and Range

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

a. (-3, 2)

b. (-2, -3)

c. (2, -3)

d. (-3, -2)

40 [Challenging] • Target: Systems involving circles • Cross-Lesson: 1.03 Average Rate of Change

In the standard (x, y) coordinate plane, a circle has its center at $(-3, 5)$ and passes through the point $(1, 8)$. What is the radius of this circle?

a. 5

b. 7

c. $\sqrt{13}$

d. 25

41 [Challenging] • Target: Circles • Cross-Lesson: 2.01 Right-Triangle Trigonometry

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

a. 1

b. 3

c. 4

d. 5

42 [Challenging] • Target: Parabolas with vertex at the origin • Cross-Lesson: 2.02 Radian Measure and Arc Length

In the circle shown below, chords AB and CD intersect at point P. If $PA = 4$, $PB = 6$, and $PC = 3$, what is the length of segment $\overline{PD}$?

a. 5

b. 7

c. 8

d. 9

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43 [Challenging] • Target: Parabolas with vertex not at the origin • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

a. $a = 3, (h, k) = (1, 1)$

b. $a = 2, (h, k) = (1, 4)$

c. $a = 3, (h, k) = (1, 4)$

d. $a = 2, (h, k) = (1, 1)$

44 [Challenging] • Target: Mixed practice: circles and parabolas together • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

In the diagram, two chords AB and CD intersect at point P inside a circle. If $PA = 6$, $PB = 8$, and $PC = 4$, what is the length of PD?

a. 6

b. 8

c. 12

d. 10

45 [Challenging] • Target: Graphing circles and parabolas • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Consider a circle with center O. Two chords AB and CD intersect at point P inside the circle. If AP = 3, PB = 8, and CP = 4, determine the length of PD. The diagram shows a circle with intersecting chords.

a. 6

b. 8

c. 10

d. 12

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